



Review

Identification of domain-relevant patents via weakly supervised deep learning[☆]

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ARTICLE INFO

Keywords:

Patent identification
Patent landscaping
Weak supervision
Deep learning

ABSTRACT

Patent identification is the process of finding patents relevant to a specific technical topic, especially in the early stages of research and development (R&D) projects. Accurately identifying relevant patents helps scientists, researchers, and industry maximize IP value, anticipate challenges, and gain insights into technological trends, competition, and future innovation opportunities. Conventional approaches, like keyword searches or retrieval based on patent classification codes, frequently result in low precision, while machine learning methods demand extensive manual annotation, posing a major bottleneck for domain-specific applications. In this work, we present a deep learning-based approach for domain-specific patent identification, with a focus on the plasma physics and cybersecurity domains. Our methodology employs a weak supervision paradigm to construct a high-quality training dataset by integrating multiple noisy labeling sources, including linguistic patterns, domain heuristics, and expert-defined rules. Using this synthesized training dataset, we fine-tune pre-trained transformer models, systematically optimizing hyperparameters to maximize performance. The resulting models can be deployed as automated patent identification systems tailored to specialized scientific and industrial contexts. We evaluate our models on previously unseen test sets using standard performance metrics. A comprehensive evaluation on unseen test set demonstrates that our approach achieves high accuracy and significantly outperforms a benchmark in-context learning approach based on large language models.

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[☆] This article is part of a Special issue entitled: 'GTM 2025's Special Issue' published in World Patent Information.

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<https://doi.org/10.1016/j.wpi.2026.102434>

Received 21 August 2025; Received in revised form 2 December 2025; Accepted 1 February 2026

Available online 5 February 2026

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1. Introduction

Patents serve as a vital source for guiding future research directions by revealing gaps that are not addressed in scientific publications [1,2]. Therefore, Patent landscaping analysis serves as an important tool for researchers in academia and industry as well as governments to assess R&D investment and innovation trends within specific scientific and technological fields. Moreover, integrating patents with scientific literature and connecting them through methods like biomedical entity extraction and relation mapping helps researchers link information across both sources [3]. This approach gives a more complete view of the research landscape, making it easier to discover patterns and insights that were previously hidden due to the isolated nature of patent data. However, one of the primary challenges that scientists face is the patent identification which is the process of efficiently identifying patents that are directly related to the area of technology they are interested in. Traditional methods for identifying patents, such as keyword searches or vector-based approaches, often suffer from limitations in the relevance and quality of the results. Even searches based on IPC (International Patent Classification) or CPC (Cooperative Patent Classification) codes often return an overwhelming number of patents with low precision. Our goal is to prioritize highly accurate and relevant patents with strong precision, which is critical during the early stages of research and development. Previous studies have focused extensively on classifying patents into predefined codes such as IPC and CPC [4], whereas this work performs targeted identification for specific, user-defined domain or topic. One widely used technique for addressing this challenge in research and development (R&D) is automatic patent landscaping which is the process of finding and analyzing patents related to a particular topic. The initial step in patent landscaping analysis involves retrieving the relevant patents from a patent databases for analysis, which is discussed in detail in the guidelines provided by the World Intellectual Property Office (WIPO) [5]. A patent document typically comprises pre-defined sections that can be roughly categorized as structured fields (metadata) such as the patent number, inventors, citations, etc. which are standardized in both semantics and format. The full-text comprises the textual fields; title, abstract, claims, and the detailed description. The latter can be divided further into several subsections such as the technical field, background, summary, embodiments, and the description of figures and drawings associated with the invention. The technical field segment, included in the patent description, is a brief section that identifies the technological domain to which the invention pertains.

Patent classification involves automatically assigning patent documents to predefined categories such as IPC or CPC codes, while patent identification focuses on finding patents that are relevant to a specific technical topic or a technological domain. A key limitation in the patent identification problem is the reliance on extensive manual annotation. The high cost and effort of labeling large training datasets create a major bottleneck, making it challenging to scale solutions and adapt models effectively to new technological domains. In this work, we introduce a new approach combining weak supervision and deep learning for patent identification. This approach designed to help researchers to automatically identify patents within a specific domain or technological area. The focus of this work is on two scientific fields: the first is plasma physics technology, which involves the use of partially or fully ionized gases known as plasma in various industrial, medical,

and scientific applications; the second is the field of cybersecurity, which is the practice of protecting digital and physical information from unauthorized access, use, disclosure, disruption, modification, or destruction to ensure confidentiality, integrity, and availability. Our approach aims to streamline the patent discovery process, making it more efficient and accessible for researchers exploring cutting-edge technologies. Weakly supervised learning is a subset of machine learning that aims to build predictive models using limited supervision. This is particularly beneficial in practical applications where labeling data is costly and time-consuming. This work utilizes the highly specific technical field section of patent documents — a concise section that defines the technological domain — as the primary input for identification process, highlighting a key methodological distinction. We employ weak supervision techniques to derive high-quality training labels from noisy or limited annotations, and we fine-tune pre-trained transformer models for automated patent identification in two domain-specific use cases: plasma physics and cybersecurity. The models are optimized through hyperparameter tuning and trained on the synthesized labeled data. We evaluate the fine-tuned models on unseen test sets, demonstrating their effectiveness in accurately identifying patent-related content. The resulting models are ultimately capable of distinguishing patents relevant to the target domains from those that are unrelated, enabling precise and efficient patent identification. Large language models (LLMs) exhibit the ability for in-context learning (ICL), which involves learning from a small number of examples provided within the context. Numerous studies have demonstrated that LLMs can successfully perform a variety of complex tasks using ICL, such as solving mathematical reasoning problems [6]. We assess LLMs for patent identification using an annotated test set focusing on the fields of plasma physics and cybersecurity.

The contributions in this work are as follows:

- We present a weakly supervised deep learning approach for patent identification, enabling accurate detection of domain-specific patents without requiring large manually annotated datasets.
- High-quality training labels are generated from noisy or limited annotations through weak supervision techniques that integrate linguistic patterns, heuristic rules, and domain-specific knowledge, effectively addressing the annotation bottleneck in the patent domain.
- We fine-tune pre-trained transformer models for two specialized scientific domains — plasma physics and cybersecurity — demonstrating the adaptability of our approach to distinct and technically complex fields, and we conduct comprehensive evaluations on previously unseen test datasets.
- The pretrained models for patent identification are evaluated against LLMs utilizing in-context learning (ICL) to assess performance on the patent identification task.

The remainder of this paper is structured as follows. Section 2 reviews the related work, while Section 3 presents our weakly supervised deep learning framework for patent identification, detailing its main components and methodological approach. Section 4 outlines the experimental design and implementation and reports the corresponding evaluation results. Section 5 discusses the findings, interprets their implications, acknowledges the study's limitations, and suggests potential directions for future research. Finally, Section 6 concludes the paper by summarizing the key outcomes of the study.

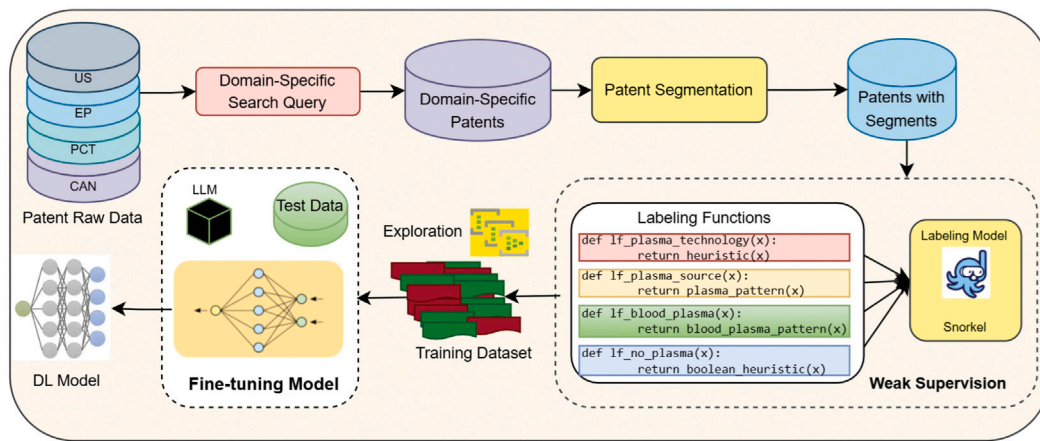


Fig. 1. The patent identification workflow.

2. Related work

Various approaches utilizing machine learning and text mining have been proposed to analyze patents and extract knowledge from patent data [7–9]. Sofean et al. [10] presented a tool named PatSeg that allows to semantically segment the text of the detailed description into predefined categories such as technical field, background, summary, etc. Substantial research on patent analysis focused on patent classification which automatically classify patent documents into existing predefined IPC or CPC classification codes [4,11]. Patent identification on the other hand is the targeted process of finding patents relevant to a specific, user-defined technological domain. Trippe [5] provided a detailed explanation in the guidelines for preparing Patent Landscape Reports, beginning with the selection of a topic and identifying the related patents following a manual or semi-automatic process. Automatically finding patents related to a specific topic was introduced by Abood et al. [12] as a patent landscaping approach. The authors developed a semi-supervised machine learning framework based on Wide-and-Deep models. This framework requires a small, representative set of patents, referred to a “seed” set as input. The algorithm leverages both patent classification codes (CPC) and citations to expand this seed set into a collection of “probably-related” patents while also identifying an “anti-seed” set of “probably-unrelated” patents. More specifically, seed patents (positive examples) form the core of the target topic, whereas anti-seed patents (negative examples) are selected based on the complementarity of the extension set. A classification model is then trained to differentiate between seed-related and anti-seed-related patents. While this approach proposed is promising, it has several notable limitations. One key issue is that the anti-seed dataset may be significantly unrelated to the target technology. For example, when identifying patents related to blockchain technology, the anti-seed set might include patents on pharmaceuticals, automobile engines, or semiconductors. Another limitation is that selecting the anti-seed data requires access to a large backend repository containing an extensive number of patents. The approach in [12] was extended in [13] by adding amount of human supervision to improve the accuracy. Choi et al. [14] proposed a deep patent landscaping model that addresses the classification problem in patent landscaping using a transformer and Diff2Vec structures. Various weakly supervised learning techniques were introduced to build predictive models using noisy data or limited supervision [15–18]. Most previous works used patent abstracts to run their experiments, but our work is based on technical field section in patent, which describe the core and secondary domains of the invention. In addition, we use weak supervision approach in order to only focus on the topic of interest by writing a precise label functions.

3. Patent identification framework

In this work, we propose a comprehensive framework for identifying patents relevant to a specific technological area with high precision and efficiency. The process begins with the formulation of a broad boolean query that integrates domain-specific keywords with patent classification codes to retrieve an extensive raw dataset. To structure the extracted data, a patent segmentation tool is employed, systematically partitioning the description texts into predefined sections including the technical field of the invention. We leverage programmatic weak supervision, utilizing heuristic-based labeling techniques to generate annotations while managing potential noise. This approach allows for scalable and efficient labeling without the need for extensive manual effort. We refine patent identification accuracy by developing a deep learning model, leveraging a pre-trained transformer architecture to capture complex linguistic patterns and domain-specific semantics. This end-to-end methodology ensures an effective and automated approach to the patent identification task. Fig. 1 shows the components, and the workflow of our framework.

3.1. Domain-specific search query

Our initial strategy for extracting patent documents involves using a carefully designed boolean query that integrates relevant patent classification codes to ensure the comprehensive retrieval of all documents related to the specific area. The extracted data serve as a candidate dataset for our framework.

3.2. Patent text segmentation

The detailed description of an invention in patent document is systematically divided into various sections, including the technical field, background, summary, embodiments, and the descriptions of figures and drawings. This work focuses particularly on the technical field section, which typically consists of concise paragraphs containing three to four sentences, though longer sections may occasionally appear. The technical field section plays a crucial role in outlining the core technological aspects of an invention, enabling patent professionals to efficiently identify related patents and gain a precise understanding of the invention’s technological domain. Moreover, the terminology used in this section often aligns with the hierarchical structure of the patent classification codes. For example, the phrase “*This invention generally relates to a plasma therapy device and more specifically to a cold plasma therapy device with enhanced safety*” corresponds to the plasma therapy domain, specifically focusing on cold plasma applications within plasma physics.

Table 1

A list of technical fields or technological areas of patents.

Patent number	Technical field
US6146599A	The present invention relates to non-thermal plasma reactors and more particularly to a dielectric barrier discharge system for decomposing hazardous compounds in a liquid or a gas.
EP1190605A4	The present invention relates to a microwave plasma burner in which a high temperature plasma is generated.
US20160331439A1	This present teachings are generally in the field of tattoo removal utilizing cold plasma-based systems and methods.
US20190053365A1	The present invention relates to an apparatus for generating plasma for beauty care or skin treatment.
WO2017092507A1	The invention relates to cybersecurity technology field in particular to an application encryption method device and application access method device.
WO2009143778A1	The present invention relates to network security technology field in particular to a support fast switching entity bidirectional authentication method.
EP4460939A1	The present application is generally in the field of quantum key distribution QKD applications and particularly relates to the calibration and or testing of QKD transmitters.

To systematically extract the technical field section from patent texts, robust and scalable segmentation techniques have been developed to partition the description section into predefined semantic sections [10]. The segmentation approach follows a hybrid methodology, integrating machine learning, rule-based algorithms, and heuristics to improve structural organization and accuracy. The description text is first extracted from the patent document and analyzed to determine whether it contains structured headings. The text is then divided into individual paragraphs, followed by a pre-processing step that removes unwanted tokens and applies stemming, a rule-based algorithm is used to identify explicit section headings, a trained machine learning model predicts the appropriate segment category for each heading. For patents lacking structured headings, heuristic methods (pattern recognition and learning-by-problem-solving techniques) are applied to classify text into relevant sections, and the final step involves detecting section boundaries and mapping text content to their respective segments. This segmentation approach achieved an accuracy of up to 94% [10]. These methods demonstrate the effectiveness of the approach in accurately structuring patent documents, ultimately enhancing patent classification and retrieval processes. Table 1 presents a selection of patents containing technical field texts.

3.3. Weak supervision

One of the most significant bottlenecks in developing machine learning applications is the high cost and effort required for manually labeling training data. To address this challenge, researchers and industry practitioners have introduced weak supervision techniques, which enable the automatic generation of labeled datasets with minimal human intervention [17,18]. A weak supervision approach eliminates the need for manually labeling large datasets, significantly saving time and effort. Instead, it generates approximate labels using heuristics, such as domain-specific rules or pattern matching techniques. Moreover, users can define weak supervision sources such as heuristics, knowledge bases, and pre-trained models which are encoded as labeling functions (LFs). These labeling functions are lightweight user-defined programs that assign labels to subsets of the data, allowing for the rapid generation of large-scale training datasets [16]. However, because labeling functions operate independently, they may generate inconsistent or contradictory labels, often introducing noisy annotations with varying degrees of accuracy. To mitigate these challenges, researchers developed labeling models like Snorkel, which aggregate and refine noisy labels to improve overall label quality. Snorkel is a weak supervision framework that allows users to train models using labeling functions, eliminating the need for manual data labeling [17]. It leverages a probabilistic modeling approach to estimate the reliability of each labeling function by analyzing its agreements and disagreements with other

functions. These estimated accuracies are then used to reweight and combine the labels produced by the functions, ultimately generating confidence-weighted probabilistic training labels. It provides an end-to-end solution for developing labeling functions, as well as training and evaluating the resulting model. The system is tailored for classification and information extraction tasks. The LFs act as black-box programs designed to process unlabeled data points and either assign a label or abstain when they lack sufficient confidence. These functions serve as a flexible mechanism to encode diverse weak supervision strategies, allowing users to incorporate domain knowledge, heuristics, or external resources into the labeling process. To enhance the reliability of the generated labels, Snorkel employs a generative modeling approach to estimate the accuracies and dependencies of different LFs. This estimation is based on analyzing the agreements and disagreements among LFs, helping to quantify their effectiveness and identify potential biases. These accuracy estimates are leveraged to reweight and aggregate the labels produced by the LFs, ensuring that more reliable sources contribute more heavily to the final labeling decision. This process results in the generation of probabilistic (confidence-weighted) training labels, which provide a more robust foundation for training downstream machine learning models [17]. This work leverages Snorkel as effective tool for the automated label generation for the technical field section of patents related to plasma physics and cybersecurity domains. Our main objective is to accurately assign relevant labels to the regarded patents based on their technological focus. To achieve this, we incorporate domain expertise from plasma physics by utilizing a curated set of plasma-related terms such as concepts and entities provided by plasma experts. In particular, our plasma experts manually curated a comprehensive list of entities and concepts related to plasma physics. These were systematically organized into multiple categories, including plasma applications, plasma targets, plasma sources, plasma media, plasma properties, plasma source properties, devices, diagnostics, materials, physical effects, and physical quantities. These terms, along with abstracts and other contextual information, are used to develop LFs within the Snorkel framework. These LFs systematically annotate patents based on the presence of domain-specific terms, enabling the creation of a high-quality labeled dataset without the need for extensive manual annotation. Additionally, we leverage the assistance of ChatGPT [19] to compile additional list of plasma technical concepts. On the other hand, we gather patterns associated with blood plasma to ensure clear differentiation from those relevant to plasma physics, preventing any unintended overlap. By analyzing these distinct plasma-related patterns, we develop a set of heuristics that serve as the foundation for constructing our labeling functions within the Snorkel framework. In particular, we develop a variety of Python functions to label data by identifying predefined terms and concepts related to plasma technology, including plasma sources, plasma medicine, plasma decontamination, and more. Some examples are described below.

If-plasma-technology verify whether the technical field section of the patent document includes any technology names associated with plasma physics.

If-blood-plasma verify if the technical field section of the patent document contains any patterns related to blood plasma.

If-non-plasma verify if the title, abstract, and technical field section of the patent document do not contain any patterns related to plasma.

In a similar effort to gather sources in plasma physics, experts in the cybersecurity domain identified relevant concepts, entities, and patterns. These resources were then used to design labeling functions, which enabled the automated creation of a training dataset tailored specifically for cybersecurity applications and some examples are described below.

If-infosec-cryptography verify whether the technical field section of the patent document includes any techniques for cryptography, including encryption, hashing, and digital signatures.

If-infosec-network-security verify whether the technical field section of the patent document includes any concept related to network security such as firewalls, intrusion detection system, wireless security, network access control, and virtual private networks.

If-no-infosec verify if the title, abstract, and technical field section of the patent document do not contain any patterns related to cybersecurity domain.

For plasma domain, three labels will be generated, plasma-physics-related, non-plasma-related, and blood-plasma-related, the latter is regarded as non-plasma-related. Furthermore, Snorkel adds a unique label called “ABSTAIN” to the set of labels whenever it is unable to determine which of the task’s labels to assign. For cybersecurity domain, two labels were created information-security-related, and non-related.

3.4. Fine-tuning transformer-based language model

Identifying patents highly correlated with specific domains, such as plasma physics, poses a challenge for the development of robust and efficient classification models with minimal pre-processing and computational overhead. To address this, we frame the task as a downstream classification problem and harness the power of transfer learning in natural language processing (NLP). Specifically, we fine-tune a pre-trained language model, allowing it to adapt to a domain-specific model and effectively classify patents with high accuracy. A variety of large language models, including GPT-3 [20], ELMO [21], and BERT [22], have been open-sourced and made freely available for use. Among these, BERT (Bidirectional Encoder Representations from Transformers) is an encoder-only model that is pre-trained using masked language modeling and next-sentence prediction. This pre-training approach allows BERT to understand the context of words in a bidirectional manner, capturing richer relationships between words and sentences. The model employs three embedding layers in its input representation, which include token embeddings, segment embeddings, and position embeddings, as detailed below.

Token Embeddings BERT model like SciBERT [23] converts the words into a fixed 768-dimensional vector representation. The tokens [CLS] and [SEP] are added to the beginning and end of the tokenized sentence.

Segment Embeddings this helps the model to understand boundaries and relationships between different segments or sentences of text, which helps to understand the context.

Position Embeddings they are the position of the word in the sentence encoded as a vector.

SciBERT Model it is a language model built on the same architecture of original BERT model [23]. It is a pre-trained language model specifically designed for scientific texts. This work fine-tunes a deep learning model using SciBERT as the pre-trained language model. It was initially pre-trained on a vast corpus of over 1.14 million research papers. One of the key innovations of SciBERT is the creation of a specialized WordPiece vocabulary called SCIVOCAB which is tailored

to the scientific domain by leveraging the SentencePiece library [24]. SciBERT has been extensively evaluated across a range of scientific tasks and datasets, consistently outperforming original BERT [22] and BioBERT [25] in all evaluated tasks, highlighting its superior ability to capture domain-specific language and context.

4. Experimental design and implementation

This section details our datasets, implementation, and experimental findings. We begin by describing our data extraction and data pre-processed incl. applying the patent text segmentation tool to extract the technical field section, applying weak supervision to construct a labeled datasets, and then fine-tuning a pre-trained DL models on the labeled datasets. We then show the evaluation results of the models on an unseen test datasets. Finally, we explore the results of an ICL approach applied to the patent identification task.

4.1. Dataset

The dataset used in this work is extracted from three different patent databases, European Patent Office (EPO), World Intellectual Property Organization (WIPO) and US Patent and Trademark Office. Using three databases to construct the training dataset ensures broader coverage, reduces source-specific biases, enriches the feature space, and enhances the quality training dataset. We started extracting patent documents as a candidate datasets by boolean search with the patent classification codes for both domains plasma physics and cybersecurity.

Plasma Physics Dataset: we use the classification code H05H which is under the heading “Plasma Technology” since each patent document has a list of classification codes. Therefore, we extract every patent which contains that classification code. 46,193 candidate patent documents were extracted, but not all of these documents are directly related to plasma technology. For this reason we adopt this work in order to identify only domain-specific related patents. We use this candidate dataset to create a training dataset. The text of each document is subjected to pre-processing removing unwanted tokens such as numbers, special symbols and xml/html tags. We consider only one member of each patent family; this means we end up with 24,513 unique patents. Blood plasma is another topic that shares the term “plasma” with plasma physics domain and both are entirely different concepts. We consider patent documents related to blood plasma in this work as a non-plasma related documents. Therefore, we extract 275 patents using the IPC code A61K0035-16 within the title “blood plasma”. A total amount of 24,788 patent documents were extracted as candidates for the process of creating a training set.

Cybersecurity Dataset: this dataset was extracted using two classification codes: G06F21, categorized under the title “Security arrangements for protecting computers, components thereof, programs or data against unauthorised activity”, and H04L9, categorized under the title “Arrangements for secret or secure communications; Network security protocols”. We extracted 96,545 unique patent family as candidates for the process of creating a training set.

4.2. Implementation and model training

This section outlines the implementation and evaluation of our framework, which includes the development of a DL models designed for automatic patent identification. We fine-tuned two separate models: one is plasma physics model and the other is cybersecurity model. We have made our implementation publicly available as open-source software, and it is accessible on GitHub [26]. Additionally, the datasets created during this study are available on request.

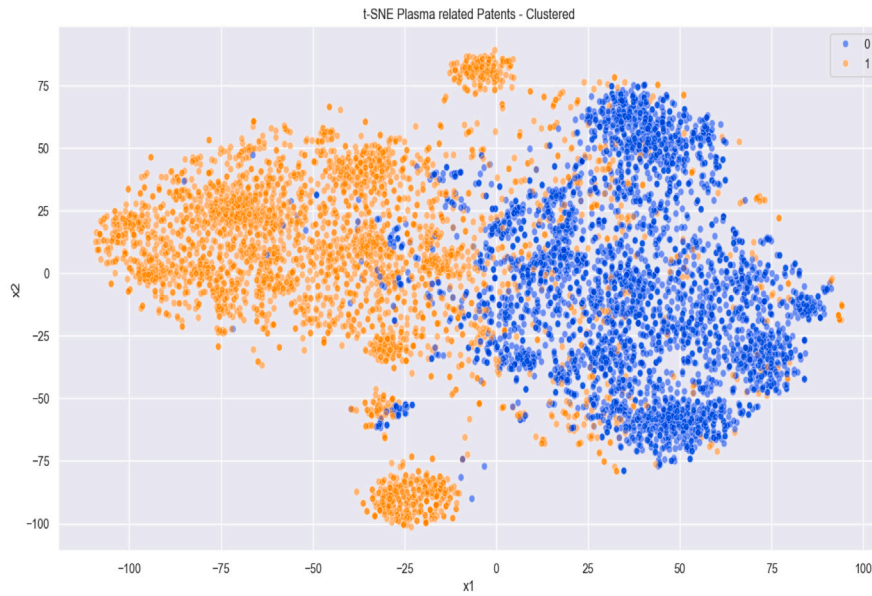


Fig. 2. The t-SNE visualization for the training dataset (Label 0 : PLASMA, Label 1: NON-PLASMA).

4.2.1. Segmentation

As outlined in Section 3.2, we utilize the segmentation tool Pat-Seg [10] to divide the detailed description texts of patents in the candidate dataset into various segments. In this paper, we focus exclusively on the technical field segments and in certain labeling tasks we also incorporate titles and abstracts in some cases. Similarly, the test dataset (Section 4.3.1) is processed in the same manner, with only the technical field section being extracted from each document. In the end, both the candidate and test datasets consist solely of technical field texts.

4.2.2. Labeling functions

To automatically label the candidate dataset extracted in Section 4.1, For the plasma physics use case, the Snorkel framework is employed to categorize the data into physics plasma-related, blood plasma-related, and no-plasma-related categories, as outlined in Section 3.3. A variety of Python labeling functions are utilized to generate probabilistic labels, facilitating the creation of labeled datasets for deep learning tasks without the need for manual annotation. The documents classified as blood plasma-related are considered as non-plasma-related patents. The final training dataset contains 6425 patent documents labeled as either “PLASMA” (3207 documents) or “NON-PLASMA” (3218 documents). For the information security use case, a set of labeling functions were developed to categorize the candidate dataset into information-security-related, and non-related. The final training dataset contains 6500 documents labeled as either INFOSEC (3000 documents) or NON-INFOSEC (3500 documents).

4.2.3. Training data exploration and evaluation

To gain an understanding of how the training dataset is structured in higher dimensions, we utilized t-SNE technique (t-distributed Stochastic Neighbor Embedding) [27], an unsupervised, non-linear dimensionality reduction technique designed for data exploration and visualization of high dimensional data. t-SNE is employed to reduce dimensionality, enabling the visualization and exploration of clustered data in a more interpretable and insightful manner. Initially, we generated Sci-BERT embeddings for each patent document in the training data using an internal embedding API based on Sentence Transformers [28]. We used K-Means for clustering the training data into two clusters namely PLASMA and NON-PLASMA. Subsequently, t-SNE was applied to visualize and explore the data as points within a multidimensional space. The t-SNE visualization allowed us to see the separation

between the clusters and figure out if there is outliers in order to refine the labeling functions and enhance training data quality. Fig. 2 shows how our training data looks like and shows a few of overlapping documents due to having similar contexts but using different techniques. For instance, patents related to “surface cleaning” using plasma are positioned near those focused on “surface cleaning” alone, as they share similar themes but differ in their methods.

To manually evaluate the quality of the training set generated using labeling functions, a sample of 20 documents was selected from the plasma physics dataset comprising 10 documents related to the domain and 10 unrelated ones. These documents were manually annotated to assess the accuracy of the labeling process. The evaluation showed that all 10 domain-related documents were correctly identified. Among the 10 unrelated documents, 9 were accurately labeled as “NON-PLASMA”, while 1 was incorrectly labeled. This indicates a high precision in identifying relevant documents, with a small margin of error in filtering out unrelated content.

Moreover, the Snorkel analysis outputs provide key insights into the coverage and overlap behavior of each labeling function, which can determine the quality of the labeling functions. Coverage measures the proportion of data points that receive at least one label, while overlap measures the proportion that are labeled by two or more (non-abstaining) LFs. Table 2 shows the coverage reports for LFs of cybersecurity domain and it interprets and contextualizes the coverage and overlaps of the labeling functions used in our experiment. The “Core Concepts” LF showed the next highest coverage at 15.14%, along with a moderate overlap rate of 6.54%, indicating that it labeled a sizeable portion of data that was also labeled by other functions. The “Access Control” LF achieved a coverage of 11.70% and an overlap of 3.50%, making it another significant contributor to the labeling process. This LF provides useful complementary supervision, though its lower overlap suggests some isolation in the subset it covers. These two LFs labeled the patents with “INFOSEC” label. The “Non-cybersecurity” LF exhibited the highest coverage at 35.02%, indicating that it labeled a substantial portion of the dataset. However, it had zero overlap with any other labeling functions, suggesting it assigned labels to a “NON-INFOSEC” label.

4.2.4. Hyperparameter optimization

We tested a broad range of values to identify the optimal hyperparameters for training the models and then selected the best-performing

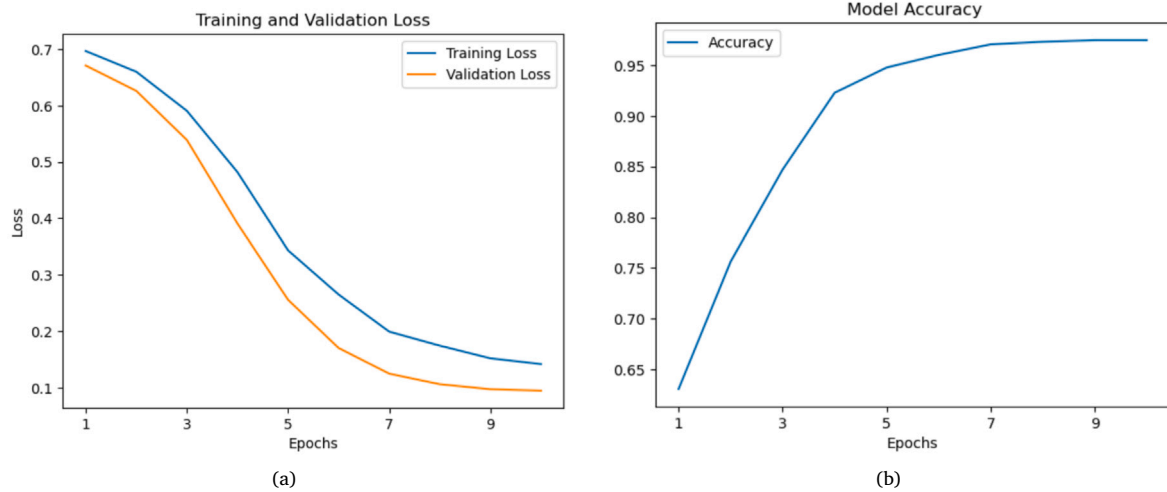


Fig. 3. Training loss and validation loss (left), and validation accuracy (right) during training for plasma physics domain.

Table 2

Coverage Report of the Labeling functions for cybersecurity Domain.

Labeling Function	Coverage	Overlaps
Core Concepts	0.151407	0.065441
Cryptography	0.018748	0.012956
Infrastructure	0.026318	0.017986
Network Security	0.021746	0.017834
Security Applications	0.003455	0.001727
Access Control	0.117010	0.034956
Cloud Security	0.001118	0.000203
Non-cybersecurity	0.350168	0.000000

ones. The training dataset in each domain was divided into a train set (70%) and a validation set (30%). The models were trained with a batch size of 16 for training and 8 for evaluation, which were the maximum sizes that could reliably fit on our GPU. We used the pytorch-transformers library [29] and the Sci-BERT model downloaded from the Hugging Face hub. This model has a hidden size of 768, 12 attention layers, and approximately 110 million parameters. Additionally, we used the scibert-scivocab-uncased variant to ensure that the tokenizer converts all the text to lowercase before fine-tuning. The model's maximum sequence length was set to 128 tokens.

4.2.5. Overfitting

Overfitting is a critical concern when training a DL model for a specific task, requiring careful consideration. To address this, an effective optimization strategy with a suitable learning rate must be implemented. To prevent overfitting, we conducted numerous experiments with varying values for learning rate, weight decay, and dropout rate. Ultimately, we determined the optimal hyperparameter values to be $5e-7$ for the learning rate, 0.0 for weight decay, and 0.3 for the dropout rate. Additionally, the dropout ratio for attention probabilities was set to 0.25. We employed Adam Torch as the optimizer, using default values of $\beta_1 = 0.9$ and $\beta_2 = 0.999$. After training the model for different numbers of epochs, we terminated the process at epoch 10 for plasma physics model and at epoch 20 for cybersecurity model, as the monitored metric showed no further improvement.

4.3. Evaluation and results

Accuracy was chosen as the primary evaluation metric for assessing the model's performance, as the training dataset is balanced. The models were fine-tuned using the training set on a GPU-enabled system

to enhance computational efficiency. During training, optimizations are used to iteratively adjust the hyperparameters at each epoch, ensuring optimal convergence. Each training and validation cycle was evaluated using several performance metrics, including training loss, validation loss, precision, accuracy, recall, and macro F1-score, providing a comprehensive evaluation of the model's effectiveness.

Model Validation and Training for Plasma Physics: Fig. 3 shows the trend of training and validation loss, illustrating a steady decline over 10 epochs, which indicates the model's progressive optimization and reduced error rates. In particular, both the training loss and validation loss decrease and stabilize at each epoch point. Additionally, Fig. 3 demonstrates the increase in accuracy values throughout the training phase, highlighting improved predictive performance as the model learns from the data. Together, these results confirm the model's ability to generalize well to validation data, making it effective for identifying related patents.

Model Validation and Training for cybersecurity: Fig. 4 presents the trend of training and validation loss over 20 epochs, showing a consistent downward trajectory that reflects the model's ongoing optimization and reduction in error. Both training and validation loss exhibit a steady decline and begin to stabilize in the later epochs, indicating convergence. Furthermore, the figure also illustrates a continuous improvement in accuracy across epochs, signifying enhanced predictive performance as the model effectively learns from the training data over time.

4.3.1. Test dataset

To evaluate the performance of the deep learning models, we assess how both models perform on unseen datasets.

Test dataset for plasma physics domain: the candidate test dataset comprising 5000 patent documents was constructed. This test dataset was randomly extracted from the Canadian patent database. For the candidate test dataset, we only considered the text from the technical field section. The annotation was carried out on the test dataset by a data scientist who has collaborated with plasma experts. At the end, the dataset included 551 patents related to plasma physics and 744 patents unrelated to plasma physics, ensuring a diverse representation of relevant and non-relevant documents (1295 in total). The dataset served as an independent benchmark to assess the model's ability to accurately classify patents based on their relevance to plasma physics.

Test dataset for cybersecurity domain: We randomly selected 1000 documents that were not part of the training dataset. Out of these, 941 documents were manually annotated with two labels: INFOSEC and NON-INFOSEC.

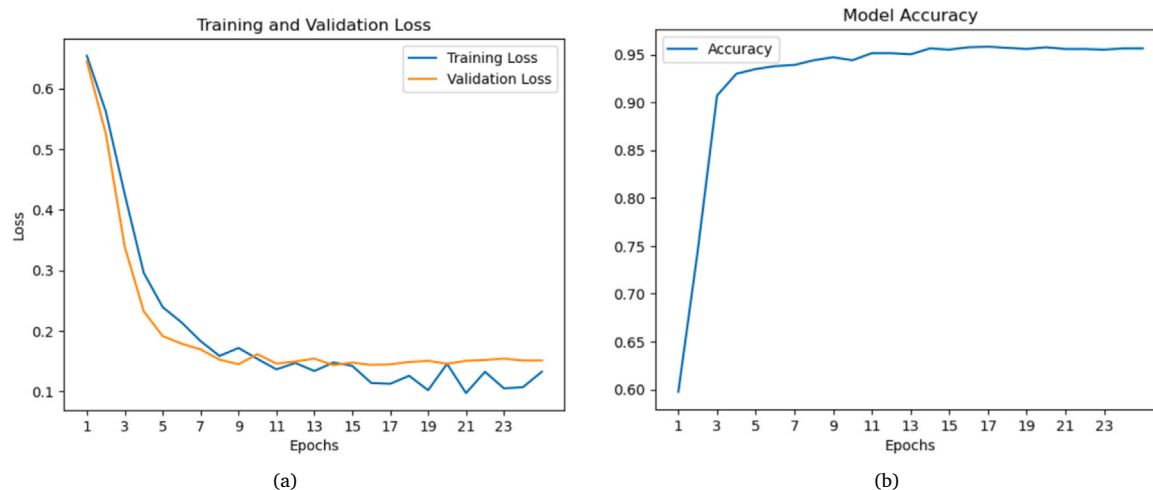


Fig. 4. Training loss and validation loss (left), and validation accuracy (right) during training for cybersecurity domain.

4.3.2. Model performance with unseen dataset

The performance of the DL models on an unseen dataset serves as a critical measure of its generalizability and robustness. Evaluating the model on previously unseen data helps to determine its effectiveness in making accurate predictions beyond the training set, ensuring that it is not overfitted to specific patterns. After applying the plasma model to this test dataset in Section 4.3.1, the model achieved an accuracy of 95%, demonstrating a strong ability to correctly classify the majority of instances. Additionally, the model's F1-measure of 95% indicates a well-balanced performance, considering both precision and recall. These results suggest that the model effectively handles the trade-off between false positives and false negatives, achieving high overall effectiveness. The high accuracy and F1-measure indicate that the model generalizes well to unseen data, making it a reliable tool for the intended task. We also evaluated the model's ability to distinguish between the two classes (e.g., PLASMA vs. NON_PLASMA) across a range of classification thresholds using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) techniques. The model achieved an AUC of 90%, indicating excellent discriminatory capability.

To assess the effectiveness of cybersecurity model, the model was tested on the unseen dataset, achieving an accuracy of 97% and an F1-score of 98%. These results demonstrate the model's strong ability to accurately classify patents related to cybersecurity, indicating high precision and recall in its predictions. These results collectively underscore the models suitability for deployment in real-world scenarios where consistent and accurate classification is essential like the patent identification task.

4.4. In-context learning for identification of plasma patents

ICL is a powerful method in natural language processing (NLP), allowing a Language model to learn and adapt based on the context provided in the input data, without requiring explicit retraining. It enables large language models (LLMs) to adapt and generate responses based on the provided examples without modifying their internal parameters. In this task, ICL is employed to identify patents related to plasma physics in our test dataset. Using this approach, the language model can understand and classify patents with minimal labeled data, relying on examples provided in the input context to make predictions. The task of ICL here involves analyzing patent documents to determine whether they pertain to plasma physics, a complex and specialized domain of physics that deals with ionized gases and their behaviors.

Recently, the LLM community has started exploring the use of "LLMs as a judge" for automated evaluation, with using sophisticated

LLMs like GPT-4 or LLaMa to evaluate other LLM results [30]. In this subsection, our goal is to evaluate how effectively text generation models can identify patents related to the field of plasma physics. To achieve this, we conducted an experiment using the Gemini model [31] in a one-shot in-context learning (ICL) setting. We created a prompt that provided the model with task-related instructions and included a detailed abstract on plasma physics, prepared by domain expert. In addition to the abstract, the prompt was further enriched with carefully selected examples illustrating whether a given text is related to plasma physics or not. These examples serve as few-shot demonstrations, guiding the model to better distinguish between relevant and irrelevant content. For the experiments, we configured the hyperparameters of the Gemini model to ensure reliable and controlled outputs. Specifically, we used the "gemini-2.0-flash-001" model, set the temperature to 0.0 to minimize randomness and encourage deterministic responses, and applied a nucleus sampling parameter top_p of 0.5 to consider a very narrow set of high-probability tokens.

The Gemini model was then tasked with classifying each document in our dataset 4.3.1 to determine whether it was related to plasma physics or not. The results were that 96 documents (8%) were misclassified. Our findings revealed that our results in this work outperformed the In-Context Learning approach, which required significantly more computational resources compared to our lightweight DL model that attained a higher accuracy of 95% with minimal resource usage. This suggests that task-specific fine-tuning on smaller models can outperform resource-intensive ICL for patent identification, particularly when efficiency is prioritized.

5. Discussion

This study proposed a deep learning approach for identifying domain-specific patents, focusing on the plasma physics and information security fields. A patent identification approach is crucial for scientists and researchers to accurately identify patents relevant to their domains while avoiding false positives. The true value of this capability lies in enhancing the worth of intellectual property (IP), helping researchers anticipate innovation challenges, and offering industries valuable insights into technological advancements and market competition. Using Snorkel for programmatic labeling and SciBERT for fine-tuned models, the plasma model achieved a high classification accuracy of 95% and an F1-score of 95% on a manually annotated test set. The system outperformed an In-Context Learning (ICL) approach. The cybersecurity model on the other hand achieved an accuracy of 97% and an F1-score of 98%. The successful application of weak

supervision highlights the viability of leveraging domain heuristics and minimal expert input to replace large-scale manual annotation, which is traditionally labor-intensive in patent analysis. By segmenting the patent texts and focusing on the “technical field” section, enabled the system to achieve both efficiency and high accuracy. To evaluate data quality and uncover mislabeled samples, t-SNE was employed for dimensionality reduction and visualization. The embedding-based clustering revealed a small number of outliers within the plasma and non-plasma categories. This visualization proved valuable for refining label functions by highlighting borderline or mislabeled instances. Despite the model’s high accuracy, a small proportion of errors was observed in the classification of test data. Additionally, the analysis of labeling functions offers quantitative insights into their performance, highlighting potential weaknesses and guiding refinements to improve coverage, consistency, and labeling accuracy. This work opens several avenues for future research. First, enhancing the robustness and coverage of labeling functions is a priority. One promising approach is to use Large Language Models (LLMs) like GPT-4 to automatically generate or refine more labeling functions by prompting them with examples and instructions, enabling semi-automated LF generation at scale. Second, incorporating few-shot learning techniques could further reduce reliance on training data and increase adaptability to other scientific domains. Another direction for future work, we plan to enhance the training dataset generation by incorporating both the technical field and the claims sections of patent documents. While the technical field often provides concise descriptions of the technological domain, it is not consistently present or clearly defined across all patents.

Another promising direction for future work involves leveraging agentic AI systems [32] to automate the creation of training datasets from patent summaries. Unlike traditional rule-based or static models, agentic AI systems can actively reason, plan, and interact with text to identify and extract the core elements of an invention. By summarizing patents in a goal-directed manner, such systems can highlight key technical concepts, innovations, and domain-specific information that are critical for accurate classification. This dynamic, context-aware summarization approach could significantly enhance the quality of training data, especially in cases where patents are lengthy, inconsistently structured, or lack explicitly defined technical fields. Ultimately, integrating agentic AI could enable more scalable and intelligent supervision in domain-specific patent analysis tasks.

6. Conclusion

This work introduced a weakly supervised deep learning approach for efficiently identifying patents in specific technological areas, demonstrated through plasma physics and cybersecurity, with potential applicability across various patent domains. By leveraging programmatic labeling functions, domain heuristics, and expert knowledge, we generated high-quality training datasets without relying on extensive manual annotation—a major bottleneck in patent analytics. We fine-tuned deep learning models on the training data to be used for patent identification. These domain-specific models achieve high accuracy and significantly outperform a benchmark in-context learning approach with large language models. The results highlight the effectiveness of integrating weak supervision with transformer-based models for scalable and precise patent identification, particularly in specialized scientific and industrial contexts where labeled data is limited. Future work will focus on enhancing the robustness of labeling functions, exploring automated LFs generation through advanced language models, expanding training signals to additional patent sections, and incorporating agentic AI techniques to further automate dataset construction. Collectively, this work lays the foundation for more efficient, accurate, and adaptable patent-identification pipelines across diverse technological fields.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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